



Bird Species Observation Analysis

Labmentix - Data Analytics Project

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Bird Species Observation Analysis

Project Overview:

This project analyzes bird species observations across two ecosystems:

-  Forest
-  Grassland
- The objective is to understand:
- Bird distribution across habitats
- Species diversity and behavior
- Environmental impact on bird activity
- Conservation priorities
- This analysis supports **biodiversity conservation, eco-tourism, and environmental planning.**

Business Objectives



- Identify habitat with more birds



- Analyze species diversity



- Study behavior patterns



- Evaluate environmental impact



- Identify at-risk species

Data Process



Data Collection (Excel sheets)



Data Cleaning & Preprocessing



Feature Engineering



EDA Analysis



SQL Analysis



Power BI Dashboard

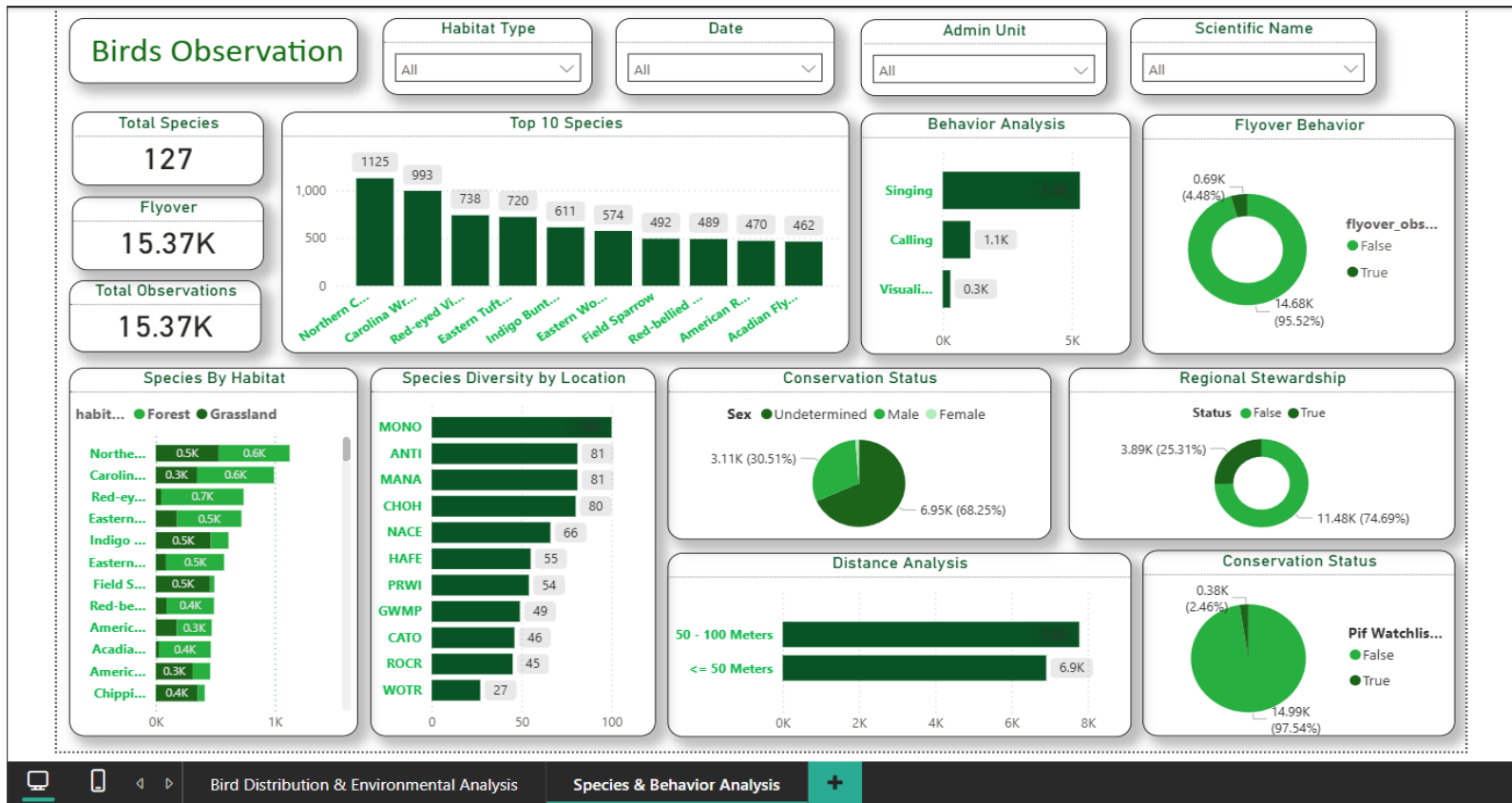
Dashboard 1

- Distribution & Environment
- Includes Habitat, Location, Time, Weather



Dashboard 2

- Species & Behavior
- Includes Species, Behavior, Distance, Conservation



Buisness Insights

Forest vs Grassland Insight:

- Forest has higher bird population and diversity.
- Grasslands support fewer but specialized species.

Biodiversity Hotspots:

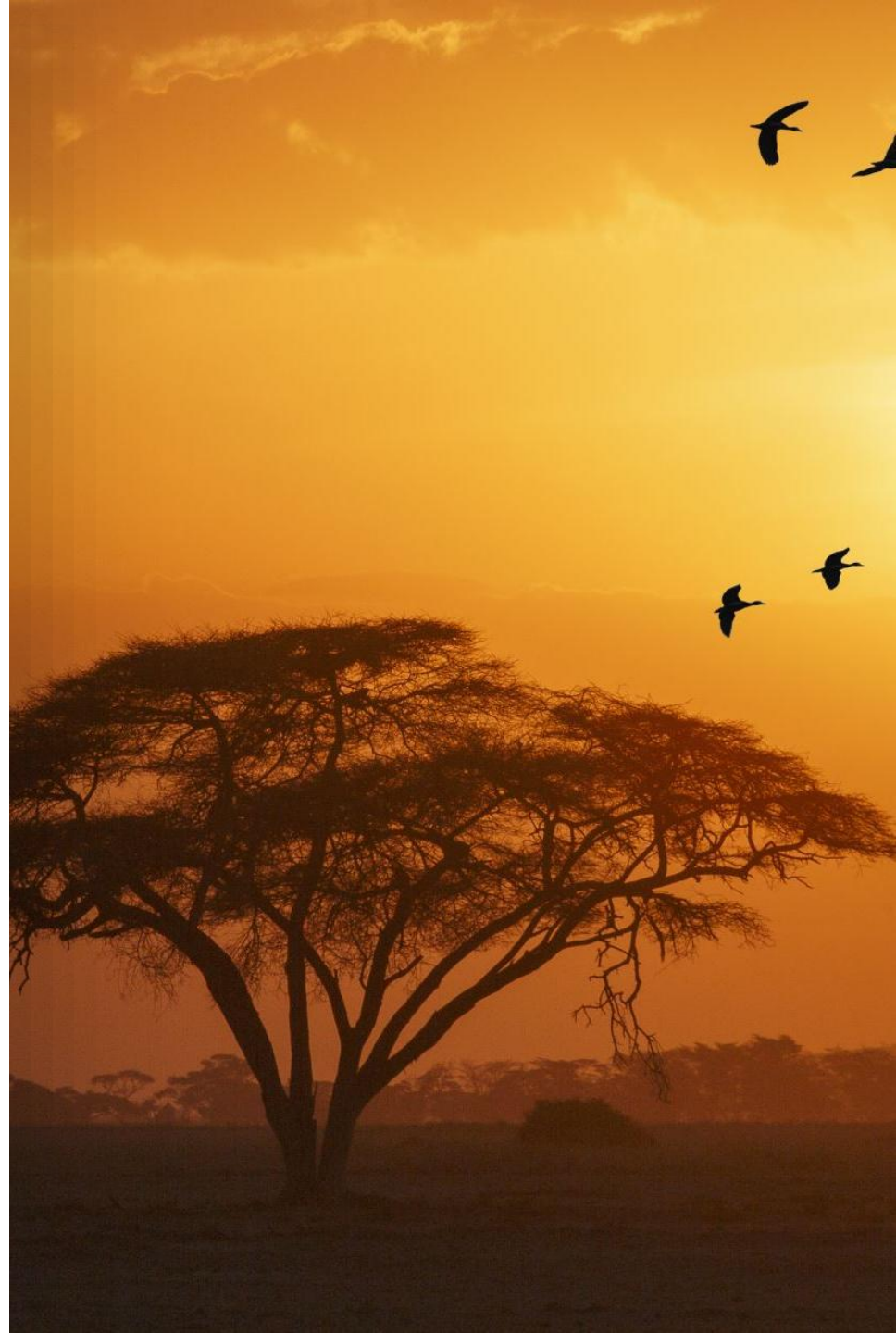
- Certain Admin Units have highest bird counts.
- Useful for conservation & tourism.

Top Species Insight:

- Few species dominate observations.
- Indicates adaptability and widespread presence.

Time Analysis:

- Birds most active in morning.
- Time Period analysis: Morning > Afternoon > Evening



Weather Impact:

- Moderate temperature increases activity.
- Extreme weather reduces sightings.

Behavior Analysis:

- Most birds identified via Singing/Calling.
- Few flyover observations recorded.

Distance & Observation:

- Birds mostly observed at shorter distances.
- Indicates localized movement.

Conservation Insights:

- Presence of watchlist species.
- Certain areas need protection.





Additional Business Insights :

Which habitat has more birds?

- **Insight:** Forests provide richer resources (food, shelter), leading to higher bird activity.

Which habitat has higher species diversity?

- **Insight:** Forest ecosystems support more biodiversity due to varied vegetation.

Which locations are biodiversity hotspots?

- **Insight:** These areas are ideal for: Conservation efforts and Bird-watching tourism

When are birds most active?

- **Insight:** Birds are more active during feeding and vocal periods.

How far are birds typically observed?

- **Insight:** Birds tend to stay within localized habitats.



Business Recommendations



Future Scope



- MACHINE LEARNING PREDICTIONS



- MAP VISUALIZATION



- REAL-TIME MONITORING SYSTEMS

Conclusion



PROJECT HELPS UNDERSTAND
BIRD ECOSYSTEM



SUPPORTS CONSERVATION &
PLANNING



Thank You